

3.6 Carboxylic Acids & Derivatives

Question Paper

Course	CIEA Level Chemistry
Section	3. Organic Chemistry
Topic	3.6 Carboxylic Acids & Derivatives
Difficulty	Hard

Time allowed: 50

Score: /36

Percentage: /100

Question 1a

Esters have important commercial uses, such as artificial flavourings in food.

They can be prepared in several ways, including through the reactions of alcohols with carboxylic acids, as well as the carboxylic acid derivatives, acid anhydrides and acyl chlorides.

Ester **S**, $\text{CH}_3\text{CH}_2\text{COOCH}_2\text{CH}(\text{CH}_3)_2$ is used in rum flavouring.

Outline how you could obtain a sample of ester **S**, beginning with a named alcohol and carboxylic acid.

Include any essential reaction conditions and write an equation for the reaction.

You do not need to include details of the separation or purification methods involved.

[6 marks]

Question 1b

Ester **S** can undergo acid or alkaline hydrolysis.

The products formed depends on the type of hydrolysis that has taken place.

Compare the **two** types of hydrolysis.

[4 marks]

Question 1c

A second ester, Ester **T** is responsible for a raspberry scent and has the molecular formula $C_5H_{10}O_2$.

Ester **T** can be produced by the reaction of an acid with a branched primary alcohol.

Identify the acid and alcohol used to prepare ester **T**.

Draw and name ester **T**.

[4 marks]

Question 2a

Lactic acid, 2-hydroxypropanoic acid, $CH_3CH(OH)COOH$ occurs naturally in sour milk and in our muscles when we take hard exercise.

Lactic acid is chiral and shows stereoisomerism.

Draw fully displayed structures of the two optical isomers of lactic acid.

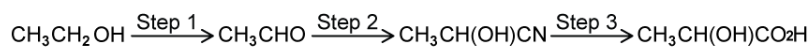
Circle the chiral carbon atom in the lactic acid molecule.

[1 mark]

Question 2b

Lactic acid may be synthesised from ethanol by the following route.

Give the reagent(s) and essential condition(s) for each step.



step 1

reagent(s)

conditions

step 2

reagent(s)

condition(s)

step 3

reagent(s)

condition(s)

[6 marks]

Question 2c

Lactic acid, $\text{CH}_3\text{CH}(\text{OH})\text{COOH}$ can be reduced by LiAlH_4 .

i)

Write an equation to show this reaction using $[\text{H}]$ to represent an atom of hydrogen from the reducing agent.

[1]

ii)

Name the organic product formed in this reaction.

[1]

[2 marks]

Question 3a

Glycolic acid is commonly used in skin care products.

The structure is shown in Fig. 3.1.

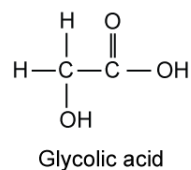


Fig 3.1

Glycolic acid is added separately to each of the three reagents below.

Complete the table to show what you would observe.

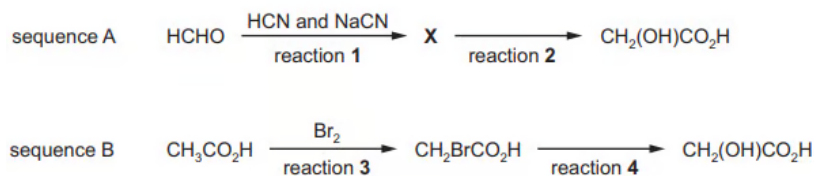
If a reaction occurs, state the functional group of glycolic acid that is responsible for the reaction.

reagent	observation with glycolic acid	does a reaction occur?	functional group
Na_2CO_3			
2,4-DNPH			
acidified $\text{Cr}_2\text{O}_7^{2-}$			

[4 marks]

Question 3b

Two reaction sequences to produce glycolic acid are shown.



i)
Draw the structure of **X**.

[1]

ii)
Name the reagent used for reaction **2**.

[1]

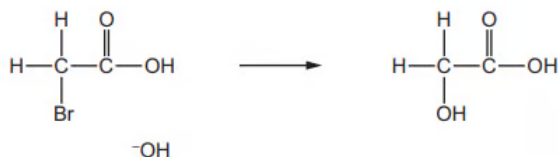
iii)
Name the mechanism for reaction **3**.

[1]

iv)
Suggest the essential condition for reaction **3**.

[1]

v)
Reaction 4 occurs via an $\text{S}_{\text{N}}2$ mechanism.
Complete the diagram for the mechanism for reaction **4**.
Include all relevant charges, partial charges, curly arrows and lone pairs.



[2]

[6 marks]

Question 3c

Glycolic acid can also be made by reacting glyoxylic acid with NaBH_4 .

i)

State the role of NaBH_4 in this reaction.

[1]

ii)

Write an equation for this reaction using molecular formulae.

Use [H] to represent NaBH_4 .

[2]

[3 marks]

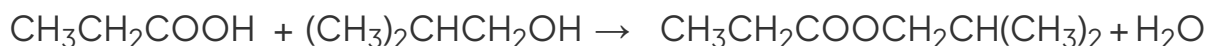
Model Answers: Hard

1a

a) To obtain $\text{CH}_3\text{CH}_2\text{COOCH}_2\text{CH}(\text{CH}_3)_2$ the reaction needs to involve:

- Propanoic acid; [1 mark]
- 2-methylpropan-1-ol; [1 mark]
- Heat; [1 mark]
- Concentrated H_2SO_4 ; [1 mark]

The equation is:



- Correct reactants; [1 mark]
- Correct products; [1 mark]

[Total: 6 marks]

- You should be able to write equations for the formation of any named ester you are given
- A good place to start is to draw the displayed formula of the ester so that you can deduce which alcohol and carboxylic acid (or derivative) it is formed from

1b

b) Compare the **two** types of hydrolysis that take place:

Similarities

- Ester **S** is heated under reflux; [1 mark]
- Both types of reaction produce the alcohol / 2-methylpropan-1-ol; [1 mark]

Differences

Any **two** from:

- Acid hydrolysis uses dilute acid / dilute H_2SO_4 whereas alkaline hydrolysis uses

dilute alkali / dilute; [1 mark]

- Acid hydrolysis produces the carboxylic acid (and the alcohol) whereas alkaline hydrolysis produces a sodium salt of the carboxylic acid / sodium propanoate; [1 mark]
- Acid hydrolysis is a reversible reaction whereas alkali hydrolysis is irreversible; [1 mark]
- Acid hydrolysis results in an equilibrium being established; [1 mark]

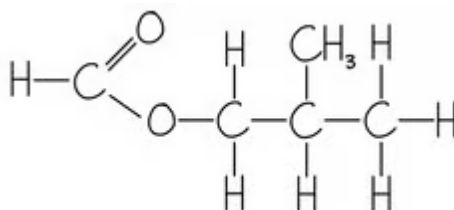
[Total: 4 marks]

- A comparison question means include similarities and differences
- When discussing the differences you must include the relevant information for each type of hydrolysis
- You will not be penalised for not identifying the names of the alcohol, carboxylic acid or sodium salt of the carboxylic acid as this is tested in part (a)
- This question is testing your knowledge of ester hydrolysis

1c

c) The acid and alcohol used to prepare ester A are:

- Methanoic acid / formic acid; [1 mark]
- 2-methylpropan-1-ol; [1 mark]
- Ester A is 2-methylpropyl methanoate; [1 mark]

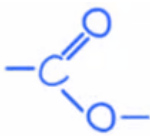
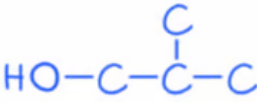
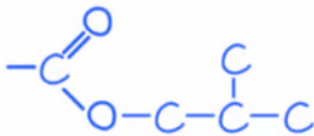
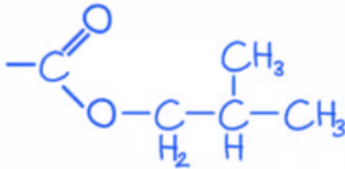
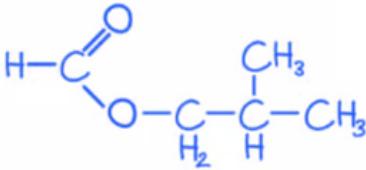
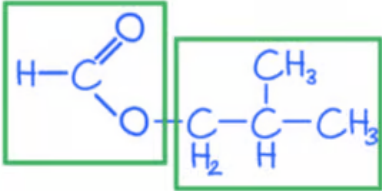


- Correct structure of ester A as shown; [1 mark]

[Total: 4 marks]

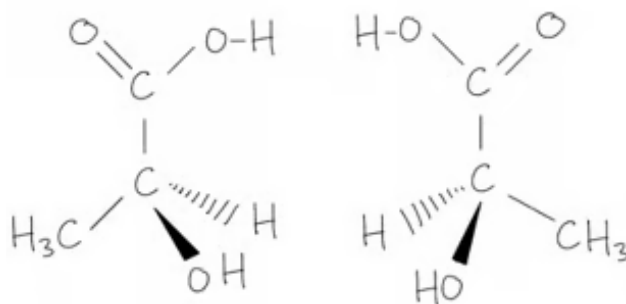
Starting from the $C_5H_{10}O_2$ formula

Draw the ester linkage / bond

(4 C and 10 H left)	
Branched primary alcohol- therefore, it must use all four carbons or it could not have a branched group	
Join the pieces together (10 H left)	
Add the hydrogens to the alcohol portion (1 H left)	
Add the hydrogen to the acid portion	
Highlight the acid and alcohol portions	
Name the acid and alcohol	Methanoic acid / 2-methylpropan-1-ol
Name the ester	2-methylpropyl methanoate

2a

a) The fully displayed structures of the two optical isomers of lactic acid are:



- Correct structures; [2 marks]
- Correct carbon atom labelled; [1 mark]

[Total: 3 marks]

- The optical isomers should be mirror images of each other
- Make sure you should all the bonds between the atoms in the carboxylic acid group as the question asks for the fully displayed structure
- You also need to use dashes and wedged lines to represent the molecule as 3D
- The chiral carbon atom is the one that has four different atoms / groups of atoms attached to it
 - The C in COOH is not chiral as it only has three groups of atoms attached

2b

b) The reagent(s) and condition(s) for each step are:

Step 1:

- Acidified potassium dichromate (VI) / $\text{Cr}_2\text{O}_7^{2-}$
AND
 $\text{H}^+ / \text{H}_2\text{SO}_4$
OR
acidified potassium permanganate / KMnO_4
AND
 $\text{H}^+ / \text{H}_2\text{SO}_4$; [1 mark]
- Distill; [1 mark]

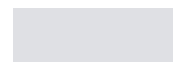
Step 2

- HCN in presence of CN^-

OR

KCN + dilute H_2SO_4 ; [1 mark]

- Room temperature; [1 mark]



OR

dilute alkali; [1 mark]

- Heat under reflux; [1 mark]

[Total: 6 marks]

- The pathway shows:
 - ethanol $\rightarrow$ ethanal $\rightarrow$ 2-hydroxypropanenitrile $\rightarrow$ 2-hydroxypropanoic acid
 - alcohol $\rightarrow$ aldehyde $\rightarrow$ nitrile $\rightarrow$ carboxylic acid
 - Higher level questions expect you to know the reagents and conditions for various multi step reaction pathways
- **Remember:** In Step 1, the potassium dichromate / potassium permanganate act as oxidising agents but can only do this by being reduced themselves, hence the need for the dilute sulfuric acid which provides H^+ ions (you will get the mark for just giving H^+)
 - The aldehyde must be distilled off as it is produced otherwise full oxidation to a carboxylic acid will occur

2c

c)

i) An equation to show this reaction using $[H]$ to represent an atom of hydrogen from the reducing agent is:

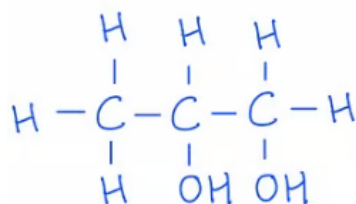
- $CH_3CH(OH)COOH + 4[H] \rightarrow CH_3CH(OH)CH_2OH + H_2O$; [1 mark]

ii) The organic produced formed in this reaction is:

- Propane-1,2-diol; [1 mark]

[Total: 2 marks]

- Carboxylic acids are reduced to their corresponding primary alcohol
- The $COOH$ group is reduced to form an OH group
- The structure of the organic compound formed is:



- There are two alcohol groups present so the compound is a diol
- These are on the first and second carbon atoms, with three carbon atoms in total hence propane-1,2-diol

3a

a) The completed table is:

reagent	observation with glycolic acid	does a reaction occur?	functional group
Na_2CO_3	effervescence / fizzing / bubbling		COOH / carboxylic acid
2,4-DNPH	no visible reaction		no group required
acidified $\text{Cr}_2\text{O}_7^{2-}$	orange to green		$-\text{OH}$ / alcohol

- 1 mark for each correct observation; [3 marks]
- 1 mark for COOH **AND** OH ; [1 mark]

[Total: 4 marks]

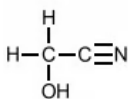
- You are expected to know the chemical tests for different functional groups
- Sodium carbonate can be used to test for the presence of the $-\text{COOH}$ group
 - When it is added to glycolic acid, a salt, carbon dioxide and water will be produced
 - The carbon dioxide can be observed as bubbles / effervescence
- 2,4-DNPH is used to test for the presence of the $\text{C}=\text{O}$ functional group found in aldehydes and ketones so will not give positive results with glycolic acid
- Acidified potassium dichromate(IV) can be used to test for the presence of the $-\text{OH}$ group is an orange oxidising agent which will turn green when it is reduced
 - Both colours must be present to score the mark

3b

b)

i) The structure of **X** is:

•



Correct structure; [1 mark]

ii) The reagent used for reaction **2** is:

- Dilute hydrochloric / sulfuric / nitric / phosphoric acid; [1 mark]

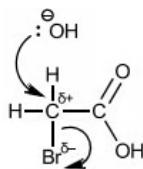
iii) The mechanism for reaction **3** is:

- Free radical substitution; [1 mark]

iv) The essential condition for reaction **3** is:

- UV / sunlight; [1 mark]

v) The diagram for the mechanism for reaction 4 is:



- Lone pair on OH⁻

AND

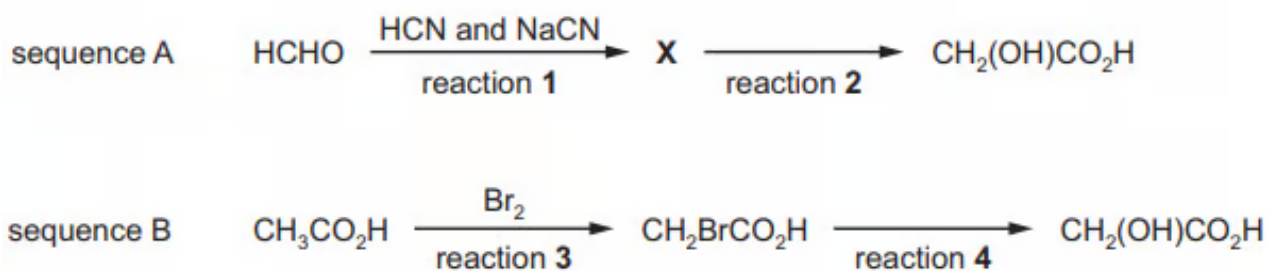
curly arrow from lone pair to C of C-Br; [1 mark]

- Correct dipole on C-Br

AND

curly arrow from bond to Br; [1 mark]

[Total: 6 marks]



- In sequence A, reaction 1, methanal undergoes nucleophilic addition to form the nitrile, 2-hydroxyethanenitrile
 - The 2-hydroxyethanenitrile then undergoes hydrolysis to form glycolic acid
 - Dilute acid is needed for this
 - 2-hydroxyethanenitrile could also undergo hydrolysis using an alkali however alkali or a named alkali is not accepted as an answer here as a carboxylate salt would be formed which would require acidification to convert it to a carboxylic acid
- In sequence B, reaction 3, hydrogen is substituted for bromine to form a halogenoalkane
 - The halogenoalkane then undergoes nucleophilic substitution to form glycolic acid
 - $\text{CH}_2\text{BrCO}_2\text{H}$ is a primary halogenoalkane so reacts via the $\text{S}_{\text{N}}2$ mechanism which is a 1 step reaction
 - Make sure your curly arrows are clear and don't forget the charge on the hydroxide ion

3c

c)

i) The role of NaBH_4 in this reaction is:

- Reducing agent / reductant; [1 mark]

ii) An equation for this reaction using molecular formulae is:



- Correct molecular formulae; [1 mark]
- Correctly balanced; [1 mark]

[Total: 3 marks]

- **Careful:** The question asks for the molecular formulae to be used so writing the

